# Eastman(TM) Turbo Oil 25

35597

| Version | Revision Date: | SDS Number: | Date of last issue: 07/13/2018 |
|---|---|---|---|
| 1.2 | 08/06/2018 | 150000094084 | Date of first issue: 09/06/2016 |
| PRD | | SDSUS / Z8 / 0260 | NCD / 0039190736 / 0001015870 |

## SECTION 1. IDENTIFICATION

Product name : Eastman(TM) Turbo Oil 25

Product code : P3435602

**Manufacturer or supplier's details**

Company name of supplier : Solutia Inc.
A Subsidiary of Eastman Chemical Company

Address : 575 Maryville Centre Drive
Saint Louis MO 63141

Emergency telephone : CHEMTREC: +1-800-424-9300, +1-703-527-3887 CCN7321

**Recommended use of the chemical and restrictions on use**

Recommended use : Lubricant

Restrictions on use : None known.

## SECTION 2. HAZARDS IDENTIFICATION

**GHS classification in accordance with 29 CFR 1910.1200**

Skin sensitization : Category 1

**GHS label elements**

Hazard pictograms :

Signal Word : Warning

Hazard Statements : H317 May cause an allergic skin reaction.

Precautionary Statements :

**Prevention:**

P261 Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray.
P272 Contaminated work clothing must not be allowed out of the workplace.
P280 Wear protective gloves.

**Response:**

P302 + P352 IF ON SKIN: Wash with plenty of soap and water.
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P363 Wash contaminated clothing before reuse.

**Disposal:**

P501 Dispose of contents/ container to an approved waste disposal plant.

**Other hazards**

None known.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

**Ingredients**

| Chemical name | CAS-No. | Concentration (% w/w) |
|---|---|---|
| N-Phenylbenzenamine reaction products with 2,4,4-trimethylpentene | 68411-46-1 | < 5 |
| Phenothiazine | 92-84-2 | < 1 |
| Tricresyl phosphate | 1330-78-5 | < 1 |
| 1,4-dihydroxy-9,10-anthracenedione | 81-64-1 | < 0.1 |

## SECTION 4. FIRST AID MEASURES

If inhaled : If breathed in, move person into fresh air.
If breathing is difficult, give oxygen.
Consult a physician if necessary.

In case of skin contact : Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.
Wash contaminated clothing before reuse.
If symptoms persist, call a physician.

In case of eye contact : In case of contact, immediately flush eyes with plenty of water for at least 15 minutes.
Get medical attention if symptoms occur.

If swallowed : Rinse mouth.
Call a physician or poison control center immediately.
Do NOT induce vomiting.
Never give anything by mouth to an unconscious person.

Most important symptoms and effects, both acute and delayed : Prolonged skin contact may defat the skin and produce dermatitis.
Contact with hot product will cause thermal burns.
Inhalation of thermal decomposition products may lead to adverse effects including pulmonary edema.
May cause an allergic skin reaction.

Notes to physician : Treat symptomatically.

## SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Water spray
Foam
Dry powder
Carbon dioxide (CO2)

| | | |
|---|---|---|
| Unsuitable extinguishing media | : | Do not use a solid water stream as it may scatter and spread fire. |
| Specific hazards during fire fighting | : | Hazardous combustion products<br>carbon dioxide, carbon monoxide<br>oxides of phosphorus |
| Further information | : | In case of fire and/or explosion do not breathe fumes.<br>Use water spray to cool unopened containers.<br>Prevent fire extinguishing water from contaminating surface water or the ground water system. |
| Special protective equipment for fire-fighters | : | Wear an approved positive pressure self-contained breathing apparatus in addition to standard fire fighting gear. |

## SECTION 6. ACCIDENTAL RELEASE MEASURES

| | | |
|---|---|---|
| Personal precautions, protective equipment and emergency procedures | : | Ventilate the area.<br>Material can create slippery conditions.<br>Use personal protective equipment.<br>Local authorities should be advised if significant spillages cannot be contained. |
| Environmental precautions | : | Avoid release to the environment. |
| Methods and materials for containment and cleaning up | : | Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13). |

## SECTION 7. HANDLING AND STORAGE

| | | |
|---|---|---|
| Advice on safe handling | : | Handle in accordance with good industrial hygiene and safety practice.<br>Wear protective gloves/ protective clothing/ eye protection/ face protection.<br>Do not get in eyes.<br>Do not get on skin or clothing.<br>Wash thoroughly after handling.<br>Contaminated work clothing must not be allowed out of the workplace.<br>Do not breathe vapors or spray mist.<br>Use only in area provided with appropriate exhaust ventilation.<br>Drain or remove substance from equipment prior to break-in or maintenance. |
| Conditions for safe storage | : | Keep containers tightly closed in a cool, well-ventilated place. |

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Ingredients with workplace control parameters**

| Ingredients | CAS-No. | Value type (Form of exposure) | Control parameters / Permissible concentration | Basis |
|---|---|---|---|---|
| Phenothiazine | 92-84-2 | TWA | 5 mg/m3 | ACGIH |
| | | TWA | 5 mg/m3 | NIOSH REL |
| | | TWA | 5 mg/m3 | OSHA P0 |

**Engineering measures** : Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level.

**Personal protective equipment**

Respiratory protection : Use respiratory protection unless adequate local exhaust ventilation is provided or exposure assessment demonstrates that exposures are within recommended exposure guidelines.

Hand protection

Material : Recommended gloves:

Material : Nitrile rubber

Remarks : Wear suitable gloves.

Eye protection : Safety glasses with side-shields
Safety goggles

Protective measures : Ensure that eye flushing systems and safety showers are located close to the working place.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : liquid

Color : amber

Odor : No data available

Odor Threshold : not determined

pH : not determined

Melting point/freezing point : -76 °F / -60 °C

Boiling point/boiling range : not determined

Flash point : 475 °F / 246 °C

| | | |
|---|---|---|
| | | Method: Cleveland open cup |
| Evaporation rate | : | not determined |
| Upper explosion limit / Upper flammability limit | : | not determined |
| Lower explosion limit / Lower flammability limit | : | not determined |
| Vapor pressure | : | not determined |
| Relative vapor density | : | not determined |
| Relative density | : | 0.995 |
| Solubility(ies) | | |
| Water solubility | : | insoluble |
| Partition coefficient: n-octanol/water | : | No data available |
| Autoignition temperature | : | not determined |
| Decomposition temperature | : | |
| | | not determined |
| Viscosity | | |
| Viscosity, dynamic | : | not determined |
| Viscosity, kinematic | : | 23 - 30 mm2/s (104 °F / 40 °C) |
| | | 4.9 - 5.4 mm2/s (212 °F / 100 °C) |
| Explosive properties | : | Not classified |
| Oxidizing properties | : | Not classified |

## SECTION 10. STABILITY AND REACTIVITY

| | | |
|---|---|---|
| Reactivity | : | None reasonably foreseeable. |
| Chemical stability | : | Stable under normal conditions. |
| Conditions to avoid | : | Keep away from sources of ignition - No smoking. |
| Incompatible materials | : | Strong oxidizing agents |
| Hazardous decomposition products | : | Emits acrid smoke and fumes when heated to decomposition. |

## SECTION 11. TOXICOLOGICAL INFORMATION

**Acute toxicity**

Not classified based on available information.

**Product:**

| | | |
|---|---|---|
| Acute oral toxicity | : | LD50 Oral (Rat): > 10,000 mg/kg |
| Acute inhalation toxicity | : | Remarks: No data available |
| Acute dermal toxicity | : | LD50 Dermal (Rabbit): > 3,160 mg/kg<br>Assessment: The substance or mixture has no acute dermal toxicity |

**Ingredients:**

**N-Phenylbenzenamine reaction products with 2,4,4-trimethylpentene:**

| | | |
|---|---|---|
| Acute oral toxicity | : | LD50 Oral (Rat): > 5,000 mg/kg |
| Acute dermal toxicity | : | LD50 Dermal (Rabbit): > 2,000 mg/kg |

**Tricresyl phosphate:**

| | | |
|---|---|---|
| Acute oral toxicity | : | LD50 Oral (Rat): > 5,000 mg/kg |
| Acute inhalation toxicity | : | LC50 (Rat): > 5.2 mg/l<br>Exposure time: 4 h |
| Acute dermal toxicity | : | LD50 Dermal (Rabbit): > 10,000 mg/kg |

**Phenothiazine:**

| | | |
|---|---|---|
| Acute oral toxicity | : | Assessment: Harmful if swallowed. |

**Skin corrosion/irritation**

Not classified based on available information.

**Product:**

| | | |
|---|---|---|
| Species | : | Rabbit |
| Assessment | : | Not classified as hazardous. |
| Result | : | slight |

**Ingredients:**

**N-Phenylbenzenamine reaction products with 2,4,4-trimethylpentene:**

| | | |
|---|---|---|
| Species | : | Rabbit |
| Assessment | : | Not classified |
| Result | : | slight |

**Tricresyl phosphate:**

| | | |
|---|---|---|
| Species | : | Rabbit |
| Exposure time | : | 24 h |

| | | |
|---|---|---|
| Assessment | : | Not classified as hazardous. |
| Result | : | Non-irritating to the skin. |

**Serious eye damage/eye irritation**

Not classified based on available information.

**Product:**

| | | |
|---|---|---|
| Species | : | Rabbit |
| Result | : | slight |
| Assessment | : | Not classified |

**Ingredients:**

**N-Phenylbenzenamine reaction products with 2,4,4-trimethylpentene:**

| | | |
|---|---|---|
| Species | : | Rabbit |
| Result | : | slight |
| Assessment | : | Not classified |

**Tricresyl phosphate:**

| | | |
|---|---|---|
| Species | : | Rabbit |
| Assessment | : | Not classified |

**Respiratory or skin sensitization**

**Skin sensitization**

May cause an allergic skin reaction.

**Respiratory sensitization**

Not classified based on available information.

**Product:**

| | | |
|---|---|---|
| Test Type | : | OECD 429: LLNA |
| Species | : | Mouse |
| Assessment | : | May cause sensitization by skin contact. |
| Result | : | sensitizing |

**Ingredients:**

**Tricresyl phosphate:**

| | | |
|---|---|---|
| Test Type | : | Skin Sensitization |
| Assessment | : | Not classified |

**Phenothiazine:**

| | | |
|---|---|---|
| Test Type | : | Skin sensitization |
| Assessment | : | May cause sensitization by skin contact. |

**Germ cell mutagenicity**

Not classified based on available information.

**Product:**

| | | |
|---|---|---|
| Genotoxicity in vitro | : | Test Type: Mutagenicity |

Result: Based on available data, the classification criteria are not met.

Genotoxicity in vivo : Test Type: Mutagenicity
Result: Based on available data, the classification criteria are not met.

**Ingredients:**

**Tricresyl phosphate:**

Genotoxicity in vitro : Test Type: various
Result: Based on available data, the classification criteria are not met.
Remarks: Not classified

Genotoxicity in vivo : Test Type: various
Result: Based on available data, the classification criteria are not met.

**Carcinogenicity**

Not classified based on available information.

**IARC** No ingredient of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.

**OSHA** No component of this product present at levels greater than or equal to 0.1% is on OSHA's list of regulated carcinogens.

**NTP** No ingredient of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP.

**Reproductive toxicity**

Not classified based on available information.

**Product:**

Effects on fertility : Remarks: No data available

Reproductive toxicity - Assessment : No toxicity to reproduction

**Ingredients:**

**Tricresyl phosphate:**

Reproductive toxicity - Assessment : May damage the unborn child. Suspected of damaging fertility.

**STOT-single exposure**

Not classified based on available information.

**Product:**

Assessment : Based on available data, the classification criteria are not met.

**Ingredients:**

**Tricresyl phosphate:**

Assessment : Based on available data, the classification criteria are not met.

**STOT-repeated exposure**

Not classified based on available information.

**Product:**

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

**Ingredients:**

**Tricresyl phosphate:**

Assessment : Based on available data, the classification criteria are not met.

**Repeated dose toxicity**

**Ingredients:**

**Tricresyl phosphate:**

Species : Rat
: 300 mg/l

**Phenothiazine:**

Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.

**Aspiration toxicity**

Not classified based on available information.

**Product:**

Not classified

**Ingredients:**

**Tricresyl phosphate:**

Not classified

**Information on likely routes of exposure**

**Product:**

Inhalation : Remarks: None known.

Skin contact : Remarks: May cause an allergic skin reaction.
Prolonged skin contact may defat the skin and produce dermatitis.

Eye contact : Remarks: None known.

Ingestion : Remarks: None known.

## SECTION 12. ECOLOGICAL INFORMATION

### Ecotoxicity

**Product:**

Toxicity to fish : LC50 (Fish):
Exposure time: 96 h
Remarks: Not classified as hazardous.
(limit of solubility in fresh water)
Read-across from a similar material

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)):
Exposure time: 48 h
Remarks: Not classified as hazardous.
(limit of solubility in fresh water)
Read-across from a similar material

Toxicity to algae : NOEC (Pseudokirchneriella subcapitata (algae)):
Exposure time: 72 h
Remarks: Not classified as hazardous.
(limit of solubility in fresh water)
Read-across from a similar material

Toxicity to fish (Chronic toxicity) : NOEC (Fish):
Remarks: Not classified as hazardous.
(limit of solubility in fresh water)
Read-across from a similar material

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC:
Remarks: Not classified as hazardous.
(limit of solubility in fresh water)
Read-across from a similar material

### Ecotoxicology Assessment

Acute aquatic toxicity : This product has no known ecotoxicological effects.

Chronic aquatic toxicity : This product has no known ecotoxicological effects.

**Ingredients:**

**Tricresyl phosphate:**

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 0.6 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0.146 mg/l
Exposure time: 48 h

M-Factor (Acute aquatic toxicity) : 1

**Phenothiazine:**

| | | |
|---|---|---|
| M-Factor (Acute aquatic toxicity) | : | 1 |
| M-Factor (Chronic aquatic toxicity) | : | 1 |

**Ecotoxicology Assessment**

| | | |
|---|---|---|
| Acute aquatic toxicity | : | Very toxic to aquatic life. |
| Chronic aquatic toxicity | : | Very toxic to aquatic life with long lasting effects. |

**Persistence and degradability**

**Product:**

| | | |
|---|---|---|
| Biodegradability | : | Remarks: Expected to be biodegradable |
| Biochemical Oxygen Demand (BOD) | : | Remarks: No data available |
| Chemical Oxygen Demand (COD) | : | Remarks: No data available |

**Bioaccumulative potential**

**Product:**

| | | |
|---|---|---|
| Bioaccumulation | : | Remarks: Mixture<br>Not applicable |

**Ingredients:**

**Tricresyl phosphate:**

| | | |
|---|---|---|
| Bioaccumulation | : | Bioconcentration factor (BCF): 2,000<br>Remarks: No information available. |
| Partition coefficient: n-octanol/water | : | Pow: 860,000<br>log Pow: 5.93 |

**Mobility in soil**

**Ingredients:**

**Tricresyl phosphate:**

| | | |
|---|---|---|
| Distribution among environmental compartments | : | log Koc: 4.31 |

**Other adverse effects**

No data available

## SECTION 13. DISPOSAL CONSIDERATIONS

**Disposal methods**

Waste from residues : Dispose of in accordance with local regulations.

## SECTION 14. TRANSPORT INFORMATION

**International Regulations**

**IATA-DGR**
Not regulated as a dangerous good

**IMDG-Code**
Not regulated as a dangerous good

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**

Not applicable for product as supplied.

**Domestic regulation**

**49 CFR**
Not regulated as a dangerous good

## SECTION 15. REGULATORY INFORMATION

**EPCRA - Emergency Planning and Community Right-to-Know**

**CERCLA Reportable Quantity**

This material does not contain any components with a CERCLA RQ.

**SARA 304 Extremely Hazardous Substances Reportable Quantity**

This material does not contain any components with a section 304 EHS RQ.

**SARA 302 Extremely Hazardous Substances Threshold Planning Quantity**

This material does not contain any components with a section 302 EHS TPQ.

**SARA 311/312 Hazards** : Respiratory or skin sensitization

**SARA 313** : This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313.

**California Prop. 65**

This product does not contain any chemicals known to the State of California to cause cancer, birth, or any other reproductive defects.

**The ingredients of this product are reported in the following inventories:**

DSL : On the inventory, or in compliance with the inventory

AICS : On the inventory, or in compliance with the inventory

IECSC : On the inventory, or in compliance with the inventory

| | | |
|---|---|---|
| TCSI | : | On the inventory, or in compliance with the inventory |
| TSCA | : | On the inventory, or in compliance with the inventory |

**TSCA list**

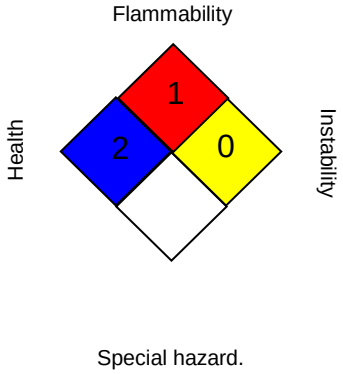
No substances are subject to a Significant New Use Rule.

No substances are subject to TSCA 12(b) export notification requirements.

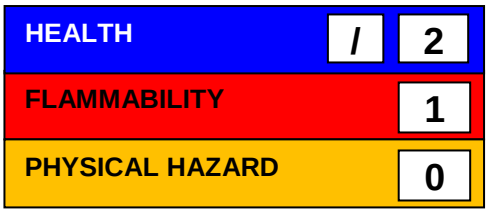
## SECTION 16. OTHER INFORMATION

### Further information

**NFPA 704:**




**HMIS® IV:**




HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. The "*" represents a chronic hazard, while the "/" represents the absence of a chronic hazard.

### Full text of other abbreviations

| | | |
|---|---|---|
| ACGIH | : | USA. ACGIH Threshold Limit Values (TLV) |
| NIOSH REL | : | USA. NIOSH Recommended Exposure Limits |
| OSHA P0 | : | USA. OSHA - TABLE Z-1 Limits for Air Contaminants - 1910.1000 |
| ACGIH / TWA | : | 8-hour, time-weighted average |
| NIOSH REL / TWA | : | Time-weighted average concentration for up to a 10-hour workday during a 40-hour workweek |
| OSHA P0 / TWA | : | 8-hour time weighted average |

AICS - Australian Inventory of Chemical Substances; ASTM - American Society for the Testing of Materials; bw - Body weight; CERCLA - Comprehensive Environmental Response, Compensation, and Liability Act; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DOT - Department of Transportation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; EHS - Extremely Hazardous Substance; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; HMIS - Hazardous Materials Identification System; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals

in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MSHA - Mine Safety and Health Administration; n.o.s. - Not Otherwise Specified; NFPA - National Fire Protection Association; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; RCRA - Resource Conservation and Recovery Act; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RQ - Reportable Quantity; SADT - Self-Accelerating Decomposition Temperature; SARA - Superfund Amendments and Reauthorization Act; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

| | | |
|---|---|---|
| Sources of key data used to compile the Material Safety Data Sheet | : | www.EastmanAviationSolutions.com |
| Revision Date | : | 08/06/2018 |

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

US / Z8